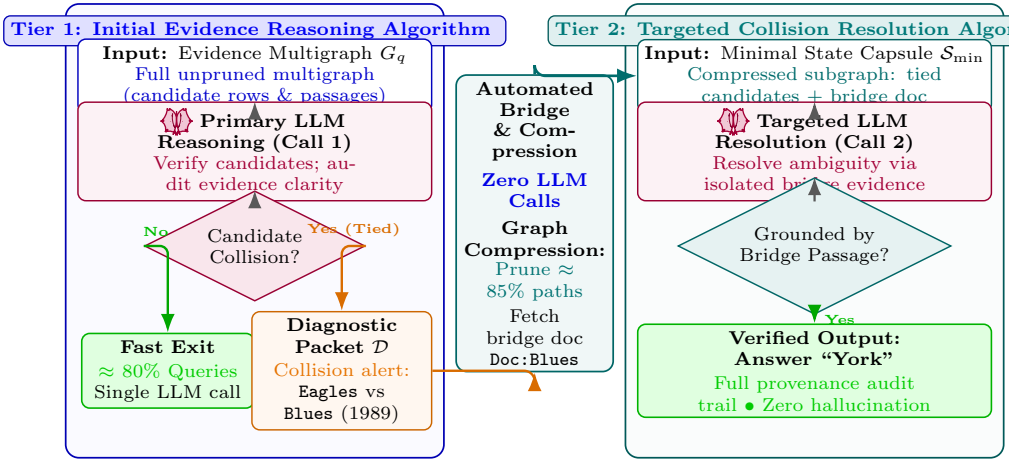

Tier 1: Initial Evidence Reasoning Algorithm
Input: Evidence Multigraph G_q
Full unpruned multigraph
(candidate rows & passages)
Primary LLM
Reasoning (Call 1)
Verify candidates; au-
dit evidence clarity
Candidate
Collision?
No
Yes (Tied)
Fast Exit
≈ 80% Queries
Single LLM call
Diagnostic
Packet $\mathcal{D}$
Collision alert:
Eagles vs
Blues (1989)
Automated
Bridge
& Com-
pression
Zero LLM
Calls
Graph
Compression:
Prune ≈
85% paths
Fetch
bridge doc
Doc:Blues
Tier 2: Targeted Collision Resolution Algorith
Input: Minimal State Capsule $\mathcal{S}_{\min}$
Compressed subgraph: tied
candidates + bridge doc
Targeted LLM
Resolution (Call 2)
Resolve ambiguity via
isolated bri e evidence
Grounded by
Bridge Passage?
Yes
Verified Output:
Answer “York”
Full provenance audit
trail • Zero hallucination